

Lab 1: Film Ranking (The Basics)

Query

Query History

```
1 WITH RankedFilms AS (  
2     SELECT c.name AS category, f.title, f.length,  
3           DENSE_RANK() OVER (PARTITION BY c.name ORDER BY f.length DESC) as rank  
4     FROM film f  
5     JOIN film_category fc ON f.film_id = fc.film_id  
6     JOIN category c ON fc.category_id = c.category_id  
7 )  
8 SELECT * FROM RankedFilms WHERE rank <= 3;
```

Data Output

Messages

Notifications

Showing rows: 1 to 64

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	category character varying (25)	title character varying (255)	length smallint	rank bigint
1	Action	Worst Banger	185	1
2	Action	Darn Forrester	185	1
3	Action	Casualties Encino	179	2
4	Action	Quest Mussolini	177	3
5	Animation	Pond Seattle	185	1
6	Animation	Gangs Pride	185	1
7	Animation	Theory Mermaid	184	2
8	Animation	Sons Interview	184	2
9	Animation	Lawless Vision	181	3
10	Children	Wrong Behavior	178	1
11	Children	Fury Murder	178	1
12	Children	Empire Malkovich	177	2

Lab 2: Growth Analysis (Time Series)

Query

Query History

```
1 WITH MonthlyRev AS (  
2     SELECT DATE_TRUNC('month', payment_date) AS month, SUM(amount) AS rev  
3     FROM payment  
4     GROUP BY DATE_TRUNC('month', payment_date)  
5 ),  
6 PrevRev AS (  
7     SELECT month, rev, LAG(rev, 1) OVER (ORDER BY month) AS prev_rev  
8     FROM MonthlyRev  
9 )  
10 SELECT month, rev, prev_rev,  
11        ((rev - prev_rev) / prev_rev) * 100 AS growth_pct  
12 FROM PrevRev;
```

Data Output

Messages

Notifications

SQL

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	month timestamp without time zone	rev numeric	prev_rev numeric	growth_pct numeric
1	2007-02-01 00:00:00	8351.84	[null]	[null]
2	2007-03-01 00:00:00	23886.56	8351.84	186.0035632866530000
3	2007-04-01 00:00:00	28559.46	23886.56	19.56288389789069669300
4	2007-05-01 00:00:00	514.18	28559.46	-98.19961581906660700200
5	2026-08-01 00:00:00	0.00	514.18	-100.00000000000000000000

Total rows: 5

Query complete 00:00:00.069

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Lab 3: Customer Lifetime Value (Running Total)

Query History

1

SELECT customer_id, payment_date, amount,
SUM(amount) OVER (
PARTITION BY customer_id ORDER BY payment_date
) AS customer_lifetime_value
FROM payment;

Data Output

Messages

Notifications

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SQL

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	customer_id smallint	payment_date timestamp without time zone	amount numeric (5,2)	customer_lifetime_value numeric
1	1	2007-02-14 23:22:38.996577	5.99	5.99
2	1	2007-02-15 16:31:19.996577	0.99	6.98
3	1	2007-02-15 19:37:12.996577	9.99	16.97
4	1	2007-02-16 13:47:23.996577	4.99	21.96
5	1	2007-02-18 07:10:14.996577	4.99	26.95
6	1	2007-02-18 12:02:25.996577	0.99	27.94
7	1	2007-02-21 04:53:11.996577	3.99	31.93
8	1	2007-03-01 07:19:30.996577	4.99	36.92
9	1	2007-03-02 14:05:18.996577	3.99	40.91
10	1	2007-03-02 16:30:04.996577	0.99	41.90
11	1	2007-03-17 11:06:20.996577	4.99	46.89
12	1	2007-03-18 02:25:55.996577	0.99	47.88
13	1	2007-03-19 08:23:42.996577	0.99	48.87
14	1	2007-03-19 12:25:20.996577	2.99	51.86
15	1	2007-03-21 22:02:23.996577	0.99	52.85

Total rows: 14597

Query complete 00:00:00.176

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Lab 4: The “Top-N” Filter Challenge

Query Query History

```

1 WITH CustSpend AS (
2     SELECT co.country, cu.customer_id, cu.first_name, SUM(p.amount) AS total_spend
3     FROM customer cu
4     JOIN payment p ON cu.customer_id = p.customer_id
5     JOIN address a ON cu.address_id = a.address_id
6     JOIN city ci ON a.city_id = ci.city_id
7     JOIN country co ON ci.country_id = co.country_id
8     GROUP BY co.country, cu.customer_id, cu.first_name
9 ),
10 RankedCust AS (
11     SELECT *, RANK() OVER (PARTITION BY country ORDER BY total_spend DESC) AS rank
12     FROM CustSpend
13 )
14 SELECT * FROM RankedCust WHERE rank = 1;

```

Data Output Messages Notifications

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	country character varying (50)	customer_id integer	first_name character varying (45)	total_spend numeric	rank bigint
1	Afghanistan	218	Vera	67.82	1
2	Algeria	176	June	151.68	1
3	American Samoa	320	Anthony	47.85	1
4	Angola	528	Claude	93.80	1
5	Anguilla	381	Bobby	99.68	1
6	Argentina	560	Jordan	129.71	1
7	Armenia	41	Stephanie	118.75	1
8	Austria	266	Nora	113.74	1
9	Azerbaijan	224	Raymond	108.75	1

Total rows: 108

Query complete 00:00:00.098

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